Derivation of every equation in `sw_sn_driver.py`

Full shallow water on a finite-amplitude meander, channel-fitted (s, n)

`numerical/dedalus_meander_full_SW`

This note derives, in order, every formula that appears in `sw_sn_driver.py`. Each section names the code symbol it produces.

Contents

1 Why (s, n), and the metric $\sigma = 1 + nC$

The channel meanders with *finite* amplitude, so its banks are not straight lines and a Chebyshev grid cannot be laid on them directly. We therefore use **channel-fitted** coordinates: s = arc length along the centreline, n = signed distance across it. In (s, n) the channel is a straight strip $n \in [-b, b]$ whatever the planform does.

Let the centreline be $\boldsymbol{r}_c(s)$ with unit tangent $\hat{\boldsymbol{t}}(s)$ and unit normal $\hat{\boldsymbol{n}}(s)$, and let $C(s)$ be its curvature. A point of the domain is

$$\boldsymbol{r}(s, n) = \boldsymbol{r}_c(s) + n\,\hat{\boldsymbol{n}}(s). \tag{1}$$

With the Frenet relation $\mathrm{d}\hat{\boldsymbol{n}}/\mathrm{d}s = +C\,\hat{\boldsymbol{t}}$ (this *defines* our sign convention: n increases *away* from the centre of curvature; equivalently the direction angle is $\theta(s) = -\int^s C\,\mathrm{d}s'$),

$$\frac{\partial \boldsymbol{r}}{\partial s} = \hat{\boldsymbol{t}} + n\frac{\mathrm{d}\hat{\boldsymbol{n}}}{\mathrm{d}s} = (1 + nC)\,\hat{\boldsymbol{t}}, \qquad \frac{\partial \boldsymbol{r}}{\partial n} = \hat{\boldsymbol{n}}, \tag{2}$$

so the coordinates are orthogonal with scale factors

$$\boxed{h_s = \sigma \equiv 1 + nC(s), \qquad h_n = 1.} \qquad (\rightarrow \texttt{sigma\_metric}, \texttt{sig})$$

σ is the local stretching of an along-channel line at offset n: the outer bank (n with the sign of C) is longer, the inner bank shorter.

Validity. $\sigma > 0$ is required or the mapping folds. Since $|n| \le b$, this is $|C|b < 1$: the bend radius must exceed the half-width. The test `tests/test_base_profiles.py` asserts $\min \sigma > 0$.

Sign convention (important). Mirroring the channel ($C \to -C$) maps the problem to itself with $n \to -n$, so the linear growth rate $\sigma(k)$ and phase speed $c(k)$ are *unchanged*; only the labelling of "outer" vs "inner" bank flips. The convention above is the one used by the solver ($\sigma = 1 + nC$); the lab-frame renderer must integrate $\theta = -\int C\,\mathrm{d}s$ to match it.

2 Shallow-water equations in (s, n)

2.1 The starting (Cartesian) system

Non-rotating depth-averaged shallow water over a bed of still-water depth H, free surface η, total depth $h = \eta + H$:

$$\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -g\nabla\eta + \boldsymbol{F}, \qquad \partial_t \eta + \nabla \cdot (h\boldsymbol{u}) = 0. \tag{3}$$

The pressure is hydrostatic, $p = \rho g(\eta - z)$, so the horizontal pressure gradient is exactly $-g\nabla\eta$; this is the $L \gg H$ (long-wave) limit. Keeping $\partial_t \eta$ is what admits **gravity waves**; the rigid-lid packages in `../` drop it and are left with the vortical branch only.

2.2 Curvilinear operators

For orthogonal coordinates with scale factors $(h_s, h_n) = (\sigma, 1)$:

$$\nabla\phi = \frac{1}{\sigma}\partial_s\phi\,\hat{\boldsymbol{t}} + \partial_n\phi\,\hat{\boldsymbol{n}}, \tag{4}$$

$$\nabla \cdot \boldsymbol{F} = \frac{1}{\sigma}\Big[\partial_s F_s + \partial_n(\sigma F_n)\Big], \tag{5}$$

$$\zeta = (\nabla \times \boldsymbol{u}) \cdot \hat{\boldsymbol{z}} = \frac{1}{\sigma}\Big[\partial_s u_n - \partial_n(\sigma u_s)\Big]. \tag{6}$$

The material acceleration acquires *metric* (centrifugal/Coriolis-like) terms from the rotation of the basis vectors. With $\mathrm{D}/\mathrm{D}t = \partial_t + (u_s/\sigma)\partial_s + u_n\partial_n$ and $\partial h_s/\partial n = C$, $\partial h_n/\partial s = 0$, the standard result is

$$a_s = \frac{\mathrm{D}u_s}{\mathrm{D}t} + \frac{C\,u_s u_n}{\sigma}, \qquad a_n = \frac{\mathrm{D}u_n}{\mathrm{D}t} - \frac{C\,u_s^2}{\sigma}. \tag{7}$$

2.3 The system solved

Substituting (5)–(7) into (3):

$$\partial_t\eta + \frac{1}{\sigma}\Big[\partial_s(hu_s) + \partial_n(\sigma h u_n)\Big] = 0, \tag{8}$$

$$\partial_t u_s + \frac{u_s}{\sigma}\partial_s u_s + u_n\partial_n u_s + \underbrace{\frac{Cu_s u_n}{\sigma}}_{\text{bend coupling}} = -\frac{g}{\sigma}\partial_s\eta - r_s u_s + \nu\nabla^2 u_s, \tag{9}$$

$$\partial_t u_n + \frac{u_s}{\sigma}\partial_s u_n + u_n\partial_n u_n \underbrace{-\frac{Cu_s^2}{\sigma}}_{\textbf{centrifugal}} = -g\,\partial_n\eta - r_n u_n + \nu\nabla^2 u_n. \tag{10}$$

The term $-Cu_s^2/\sigma$ is the whole reason for going to (s, n): it is the centrifugal acceleration of water rounding a bend, and it is what tilts the free surface up against the outer bank (§4). No straight-channel model contains it.

3 Non-dimensionalisation: where the Froude number enters

Scale lengths by the half-width b, speeds by the centreline speed $U_c = U_0 + \Delta$, time by b/U_c, and depths/η by the mean depth H_0. The only place g survives is the pressure gradient, where it appears as

$$\frac{gH_0}{U_c^2} = \frac{1}{F^2}, \qquad \boxed{F \equiv \frac{U_c}{\sqrt{gH_0}}, \qquad g_{\text{eff}} = \frac{1}{F^2}} \qquad (\to \texttt{g\_eff})$$

so the non-dimensional gravity-wave speed is $\sqrt{g_{\text{eff}}H_0} = 1/F$.

Why F must be specified, and why one value proves nothing. F is an *input* (a property of the river), exactly like a Reynolds number: without it the system (8)–(10) is not defined. At any F *both* families exist — gravity waves at speed $1/F$ and the vortical branch at $O(U)$. Which one the bank couples to is decided by *measurement*: compare c_{meander} with $1/F$, and above all *vary* F. If the meander rode the gravity branch, tripling $1/F$ would move σ, c. It does not (`sweep_dispersion.py` → `fig01`), which is the evidence that the meander is the vortical/Rossby wave.

4 The base state

4.1 Parabolic jet and the constant channel-β

$$\boxed{\bar{U}_s(n) = U_0 + \Delta\Big(1 - \frac{n^2}{b^2}\Big)} \qquad (\to \texttt{ubar\_s})$$

so $\bar{U}_s(0) = U_0 + \Delta$ (centre) and $\bar{U}_s(\pm b) = U_0$ (bank edge), and

$$\partial_n\bar{U}_s = -\frac{2\Delta n}{b^2} \quad (\to \texttt{ubar\_s\_n}), \qquad \partial_n^2\bar{U}_s = -\frac{2\Delta}{b^2} = \text{const} \quad (\to \texttt{ubar\_s\_nn}). \tag{11}$$

This constant is the load-bearing property. From (6) with $u_n = 0$ and (for the moment) $C = 0$, the base vorticity is $\bar{\zeta} = -\partial_n \bar{U}_s$, hence

$$\boxed{\frac{\partial \bar{\zeta}}{\partial n} = -\partial_n^2 \bar{U}_s = \frac{2\Delta}{b^2} = \text{constant}} \tag{12}$$

— a uniform cross-channel gradient of background vorticity, which is why the jet is taken parabolic rather than, say, sinusoidal. It is tempting to call this a channel β and conclude that a *vortical (Rossby-type) wave* is thereby possible; that inference does **not** follow, and we do not make it. A planetary β supports discrete Rossby modes because the domain is unbounded or the PV profile has an edge; a smooth, single-signed $\partial_n \bar{q}$ between two walls has no PV edge, and by the classical result of Case (1960) it supports only a *continuous* spectrum — sheared disturbances that decay algebraically, not an exponentially growing normal mode. Discrete modes require a PV discontinuity or extremum (Bretherton 1966; Rayleigh's inflexion criterion), which (12) explicitly excludes. What Δ does give us is a clean control knob: Δ is the shear, so setting $\Delta = 0$ removes the "β" entirely and tests directly whether the growing meander needs it (§12). (For $C \neq 0$ the same calculation gives $\bar{\zeta} = -\partial_n \bar{U}_s - C\bar{U}_s/\sigma$, a curvature correction.)

4.2 Base curvature = the finite background meander

$$\boxed{\bar{C}(s) = \bar{C}_{\text{amp}} \cos(k_m s), \qquad \bar{C}_{\text{amp}} = A_{\text{bank}} k_m^2 \text{ (default)}} \qquad (\to \texttt{cbar})$$

For a centreline of lateral amplitude A_{bank} and wavenumber k_m, the small-slope curvature is $-\partial_s^2(A_{\text{bank}} \cos k_m s) = A_{\text{bank}} k_m^2 \cos k_m s$; `Cbar_amp` overrides this to set a genuinely tight bend directly. A_{bank} **is the fixed background channel shape — it is *not* the perturbation seed A_0 of §10, and it does not evolve.**

4.3 Superelevation

For a steady base with $\bar{U}_n = 0$ and $\partial_s(\cdot) = 0$, the n-momentum (10) reduces to a balance between the centrifugal term and the pressure gradient:

$$\boxed{g\, \partial_n \bar{\eta} = \frac{\bar{C}\, \bar{U}_s^2}{\bar{\sigma}}} \qquad (\to \texttt{etabar})$$

integrated as $\bar{\eta}(s,n) = \dfrac{\bar{C}(s)}{g} \displaystyle\int_0^n \frac{\bar{U}_s^2(n')}{1 + n'\bar{C}(s)}\, \mathrm{d}n'$ (zero on the centreline). The water surface tilts *up towards the outer bank* — the superelevation. Note $\bar{C} \to 0 \Rightarrow \bar{\eta} \to 0$ (asserted in the test).

4.4 Discharge conservation

The base continuity (8) with $\bar{U}_n = 0$ requires $\partial_s(\bar{h}\bar{U}_s) = 0$: the transport is the same through every cross-section. With a cross-channel-only bed this is automatic ($\bar{U}_s = \bar{U}_s(n)$).

What sustains the base. A parallel jet with bottom drag is not a steady solution unless something drives it (the downhill slope). As in every frozen-jet stability analysis we *prescribe* $\bar{U}_s$ and absorb the residual into an implicit steady forcing; it cancels from the perturbation equations below.

5 Linearisation

Write $u_s = \bar{U}_s + u_s'$, $u_n = u_n'$, $\eta = \bar{\eta} + \eta'$, $h = \bar{h} + \eta'$, $C = \bar{C} + C'$, and drop products of primes.

5.1 Continuity

$\partial_s(hu_s) = \partial_s(\bar{h}\bar{U}_s) + \bar{h}\partial_s u'_s + \bar{U}_s\partial_s\eta'$, where the first term vanishes (discharge), and $\partial_n(\sigma h u_n) \to \partial_n(\bar{\sigma}\bar{h}\, u'_n)$. Hence

$$\bar{\sigma}\,\partial_t\eta' + \bar{h}\,\partial_s u'_s + \bar{U}_s\,\partial_s\eta' + \partial_n\big(\bar{\sigma}\bar{h}\,u'_n\big) = 0, \qquad (\to \text{eq. 1 in } \texttt{build\_ivp\_SW})$$

already multiplied by $\bar{\sigma}$ (§6).

5.2 s-momentum

$\frac{u_s}{\sigma}\partial_s u_s \to \frac{\bar{U}_s}{\bar{\sigma}}\partial_s u'_s$; $u_n\partial_n u_s \to u'_n\partial_n\bar{U}_s$; $\frac{C u_s u_n}{\sigma} \to \frac{\bar{C}\bar{U}_s u'_n}{\bar{\sigma}}$. Multiplying by $\bar{\sigma}$ collects the two u'_n terms into one coefficient:

$$\bar{\sigma}\partial_t u'_s + \bar{U}_s\partial_s u'_s + \underbrace{\left(\bar{\sigma}\,\partial_n\bar{U}_s + \bar{C}\,\bar{U}_s\right)}_{\to\,\texttt{coefUn}} u'_n + g\,\partial_s\eta' + \bar{\sigma} r_s u'_s - \bar{\sigma}\nu\nabla^2 u'_s = 0. \qquad (\to \text{eq. 2})$$

The u'_n coefficient is the cross-channel advection of background momentum — proportional to the base vorticity — and is the term that carries the Rossby-type restoring.

5.3 n-momentum and the curvature feedback

Expanding $-Cu_s^2/\sigma$ with $C = \bar{C} + C'$, $\sigma = \bar{\sigma} + nC'$:

$$-\frac{Cu_s^2}{\sigma} \simeq \underbrace{-\frac{\bar{C}\bar{U}_s^2}{\bar{\sigma}}}_{\text{base}} \underbrace{-\frac{2\bar{C}\bar{U}_s u'_s}{\bar{\sigma}}}_{\to\,\texttt{twoCbUb}} \underbrace{-\frac{C'\bar{U}_s^2}{\bar{\sigma}}}_{\textbf{new bend}} + \frac{nC'\,\bar{C}\bar{U}_s^2}{\bar{\sigma}^2}. \tag{13}$$

The perturbed pressure term contributes $-\bar{\sigma}g\partial_n\eta' - nC'g\partial_n\bar{\eta}$, and using the base balance ($\to$ `etabar`), $g\partial_n\bar{\eta} = \bar{C}\bar{U}_s^2/\bar{\sigma}$, the last piece is $-nC'\bar{C}\bar{U}_s^2/\bar{\sigma}$. Adding it to the $+nC'\bar{C}\bar{U}_s^2/\bar{\sigma}^2$ above and using $\bar{\sigma} - n\bar{C} = 1$ leaves the remarkably clean forcing

$$\boxed{n\text{-momentum forcing} = \frac{C'\,\bar{U}_s^2}{\bar{\sigma}} \;\simeq\; C'\bar{U}_s^2.} \tag{14}$$

So the final equation, with the forcing moved to the left and $C' = -\partial_s^2\zeta_c$ (§9):

$$\bar{\sigma}\partial_t u'_n + \bar{U}_s\partial_s u'_n - 2\bar{C}\bar{U}_s u'_s + \bar{\sigma}g\partial_n\eta' + \bar{\sigma}r_n u'_n - \bar{\sigma}\nu\nabla^2 u'_n + \underbrace{\bar{U}_s^2\,\partial_s^2\zeta_c}_{\text{bend}\,\to\,\text{flow}} = 0. \qquad (\to \text{eq. 3})$$

This is the coupling that closes the meander instability: a bend creates a centrifugal forcing, which drives the flow, which erodes the bank, which deepens the bend.

6 Why every equation is multiplied by σ

The raw equations carry $1/\sigma = 1/(1+nC)$, a *rational* function of n. Dedalus builds such a coefficient as a dense operator (its documentation warns about exactly this), destroying the banded structure. Multiplying each equation through by σ removes every $1/\sigma$ and leaves coefficients that are polynomial in n (products of σ, $\bar{h}$, $\bar{U}_s$), so the matrices stay banded. Multiplying an equation by a nonzero function is an exact, solution-preserving row operation. The one residual $1/\bar{\sigma}$ (in the C' forcing) is dropped at $O(n\bar{C}\,C')$, i.e. second order in the perturbation amplitude *and* the base curvature.

7 Bottom friction: the factor of two, and the free-surface term

Quadratic (Chézy) drag is $\boldsymbol{F} = -\dfrac{C_f}{h}|\boldsymbol{u}|\boldsymbol{u}$. Two things must be linearised: the *velocity* in $|\boldsymbol{u}|\boldsymbol{u}$, and the *depth* in $1/h$. With $\boldsymbol{u} = (\bar{U}_s + u_s',\, u_n')$, $h = \bar{h} + \eta'$ and $|\boldsymbol{u}| \simeq \bar{U}_s + u_s'$:

$$-\frac{C_f|\boldsymbol{u}|u_s}{h} \simeq \underbrace{-\frac{C_f\bar{U}_s^2}{\bar{h}}}_{\text{base}}\underbrace{-\frac{2C_f\bar{U}_s}{\bar{h}}u_s'}_{r_s\ (\to\texttt{rs\_a})}\underbrace{+\frac{C_f\bar{U}_s^2}{\bar{h}^2}\eta'}_{\textbf{superelevation drag}\ (\to\texttt{reta\_a})}, \qquad -\frac{C_f|\boldsymbol{u}|u_n}{h} \simeq -\underbrace{\frac{C_f\bar{U}_s}{\bar{h}}}_{r_n}u_n'. \quad (15)$$

The drag is **anisotropic**: streamwise perturbations are damped twice as fast as cross-stream ones, because they perturb $|\boldsymbol{u}|$ as well as the direction.

The last term is Ikeda–Parker–Sawai's *superelevation-drag* term [1]: where the surface is raised the column is deeper, the drag $\propto 1/h$ is weaker, and the flow there accelerates. **It is the only route by which the free surface drives bend flow, and hence the only place the Froude number enters classical bend theory.** Omitting it (as an earlier version of this code did) removes the physical mechanism this package exists to add, and makes any Froude sensitivity test vacuous. The base part $C_f\bar{U}_s^2/\bar{h}$ belongs to the implicit driving balance and cancels.

8 Boundary conditions, and why there are four tau terms

The banks sit at fixed $n = \pm b$ in channel coordinates. Water cannot cross a solid bank, so

$$\boxed{u_n'(\pm b) = 0} \qquad \text{(no penetration).} \quad (16)$$

This assumes *nothing* about the interior divergence: $\nabla \cdot \boldsymbol{u} \neq 0$ everywhere inside (that is what the free surface is for). In particular the bank is *not* treated as a material streamline — that would be a rigid-lid (non-divergent) assumption and is inconsistent with (8).

Lateral eddy viscosity $\nu\nabla^2$ makes each momentum equation second order in n, which (i) is physically reasonable for a river, (ii) makes the parabolic jet an actual steady solution, and (iii) gives a standard tau closure. A second-order operator needs one boundary condition per wall per component, so with free slip

$$\partial_n u_s'(\pm b) = 0 \qquad \text{(free slip),} \quad (17)$$

the tau budget is 2 per viscous velocity component:

$$\underbrace{2}_{u_s'} + \underbrace{2}_{u_n'} + \underbrace{0}_{\eta'} = 4 \qquad (\to\ \texttt{t\_us1,t\_us2,t\_un1,t\_un2}),$$

η' needing none because its equation contains no second n-derivative. Counting unknowns and equations: $\{u_s', u_n', \eta', \zeta_c\} + 4$ tau $= 8$ and 3 PDE $+$ 2 no-penetration $+$ 2 free-slip $+$ 1 erosion $= 8$. The tau terms are lifted onto the last two Chebyshev modes, `lift(t,-1)+lift(t,-2)` on `ybasis.derivative_basis(2)`.

9 Erodible bank: the meander equation

The bank is a sediment boundary, not a material line: it recedes because the flow erodes it. Ikeda's closure makes the migration rate proportional to the *near-bank excess longitudinal velocity.* For a **bend** the outer bank is fast and the inner bank slow, so it is the **antisymmetric** part that moves the centreline:

$$\boxed{\partial_t\zeta_c = E \cdot \tfrac{1}{2}\left[u_s'(+b) - u_s'(-b)\right], \qquad E = \varepsilon\, U_0} \quad (18)$$

where $\zeta_c(s,t)$ is the lateral offset of the centreline (constant width).

Why antisymmetric — and a correction. The antisymmetric combination is not an extra modelling choice: it follows from taking each bank's *outward* normal. The $+b$ bank's outward normal is $+\hat{\boldsymbol{n}}$ and the $-b$ bank's is $-\hat{\boldsymbol{n}}$, so outward erosion of each bank is $\partial_t \xi_+ = +E u_s'(+b)$ and $\partial_t \xi_- = -E u_s'(-b)$, whence the centreline $\frac{1}{2}(\xi_+ + \xi_-)$ obeys exactly (18), and the width mode $\frac{1}{2}(\xi_+ - \xi_-)$ obeys $\partial_t(\cdot) = \frac{E}{2}[u_s'(+b) + u_s'(-b)]$, which vanishes identically because the response u_s' is *odd* in n. *Correction to an earlier version of this note:* it claimed that "independent banks" would instead drive the centreline through the *symmetric* part and give $\sigma < 0$. That was wrong — it used $+\hat{\boldsymbol{n}}$ for both banks. The two formulations are **identical**, not opposed; the earlier $\sigma < 0$ came from that sign error killing the feedback, not from parity.

Closing the loop. The migrating centreline changes the curvature,

$$C'(s,t) = -\partial_s^2 \zeta_c, \tag{19}$$

which re-enters the n-momentum as the forcing derived in §5.3. The cycle

$$\zeta_c \;\longrightarrow\; C' \;\longrightarrow\; \text{flow} \;\longrightarrow\; u_s'(\pm b) \;\longrightarrow\; \partial_t \zeta_c$$

is the meander instability. Its growth rate is what the IVP measures.

10 Seeding, and why the seed must be a domain mode

The base state is an exact steady solution: started there, nothing ever happens. Stability analysis therefore *requires* a perturbation, and its growth rate is the answer. Because the dynamics are linear, *any* perturbation is admissible.

Broadband seeding (the default). We seed the centreline with all resolvable modes at once, equal amplitude and fixed pseudo-random phases:

$$\zeta_c(s,0) = A_0 \sum_{m=1}^{N_s/2-1} \cos\bigl(k_m s + \phi_m\bigr), \qquad k_m = \frac{2\pi m}{L_s}, \qquad u_s' = u_n' = \eta' = 0. \tag{20}$$

For a *straight* base channel the coefficients are s-independent, so the s-Fourier modes decouple exactly: each k_m then evolves on its own and a **single run contains the entire dispersion relation**, recovered by demodulating each mode ($\rightarrow$ `per_mode_dispersion`). This is both cheaper and safer than one run per wavelength — in particular it removes any question of whether the seeded wavenumber is commensurate with the domain. Runs are accordingly identified by their physics — bed H, initial bank sinuosity $\bar{C}_{\text{amp}} b$, friction C_f, and the base speeds U_0, Δ ($\rightarrow$ `run_tag`) — never by a seeded wavelength. Δ belongs in that label even though it reads as a mere "speed": by (12) it *is* the cross-channel vorticity gradient, so $\Delta = 0$ (plug) and $\Delta < 0$ (reversed shear) are the two sharpest experiments in the study, and omitting it would let them collide. At finite bank sinuosity the coefficients are s-periodic, the modes couple (Floquet), and the per-mode numbers are Bloch quantities rather than eigenvalues.

Convergence gate. A fitted growth rate is meaningless unless the mode actually grew: we record the e-foldings achieved over the fit window and mark any mode with fewer than three as not converged. (Without this gate, slowly-evolving modes return fitted rates whose *sign* depends on the fit window.)

A_0 **is a free constant.** The system is linear, so if $(u'_s, u'_n, \eta', \zeta_c)$ solves it then so does $\lambda\times$ it. Only σ, c and the *ratios* between fields are physical; the overall size is set by whatever initial disturbance one assumes. This is why the movie may fix the overall constant once (final meander $= 0.5b$) and apply it to every frame and every field — that is a re-choice of A_0, not a distortion.

11 Measuring σ and c

The centreline is demodulated onto its seeded mode, $a(t) = \widehat{\zeta_c}(m, t)$, and

$$\sigma = \frac{\mathrm{d}}{\mathrm{d}t} \ln |a|, \qquad c = -\frac{1}{k}\frac{\mathrm{d}}{\mathrm{d}t} \arg a, \qquad (\rightarrow \texttt{measure\_sigma\_c})$$

both fitted over the last two thirds of the run (dropping the initial adjustment, during which the flow spins up from rest). $\sigma > 0$ is an unstable meander; $c > 0$ means the bend train migrates downstream.

12 What the model does *not* contain, and what follows

Strictly linear perturbation dynamics (no $\boldsymbol{u}'\cdot\nabla\boldsymbol{u}'$); no rotation ($f = 0$, a river); no vertical structure, hence no helical secondary flow (depth-averaged away, i.e. this is the $A = 0$ limit of [1], whose $A \approx 2.89$ is their *dominant* bend driver); the base state is frozen (its steady residual is an implicit forcing); and the lateral viscosity uses the flat rather than the exact curvilinear vector Laplacian — a wall closure and short-wave regulariser.

Three consequences must be stated plainly, because they bound what any result here can mean.

(i) No sediment, therefore not the *observed* wavelength selection. The bed H is prescribed and only the banks move. Meander wavelength selection is understood as a resonance between the forced bend flow and the *free alternate-bar mode of the sediment bed* [3, 4, 6]. With the bed frozen that free mode does not exist. The driving is short-wave biased — the $C' = -\partial_s^2\zeta_c$ forcing scales as k^2 while the response scales as k — so a preferred wavenumber can only come from whatever damps short waves.

It turns out something does, and it is not the artificial viscosity: measurements in §14 give a k^2 damping coefficient about twenty-six times ν, set by the cross-channel shear, with $k_{\text{peak}}\Delta \simeq 1.10$. So the model does select a scale — by shear, not by bar–bend resonance. It is the wrong mechanism for a river, and it disappears as $\Delta \to 0$ (whereupon the peak retreats to the genuine viscous cutoff), so no comparison with the observed $\lambda \approx 10W$ [14] is warranted. The point of this subsection stands; only the reason does not.

(ii) The erodibility is not in the geomorphic regime. Field calibration gives $E/U \sim 10^{-8}$ [11, 12]; the default here is $O(10^{-1})$. Classical theories take $E \to 0$, which is what makes the flow quasi-steady and justifies imposing $u'_n(\pm b) = 0$ at a *fixed* wall. At $O(1)$ erodibility the neglected wall-normal velocity $Eu'_s(\pm b)$ is not small compared with the retained u'_n, so the boundary condition is inconsistent at leading order, and σ, c are properties of a strongly coupled flow–bank oscillator rather than of a river.

(iii) Froude insensitivity is weak evidence by construction. Writing $P \equiv g\eta$, the momentum equations contain no F at all and F survives only as F^2 multiplying the time-and-advection derivative of P in continuity — i.e. purely as a divergence (compressibility) parameter. Any slow, balanced mode is therefore F-insensitive as a matter of scaling, and the observed insensitivity *confirms* the rigid-lid reduction rather than discriminating between wave families. Distinguishing a free vortical wave from a forced boundary mode requires the diagnostics of §13, not a Froude sweep.

13 Classifying the mode (what the diagnostics mean)

`classify_mode()` reports, from the perturbation fields:

$\|\delta'\|/\|\zeta'\|$ divergent vs rotational: $\to 0$ means balanced (not a gravity wave),

$\|q'\|/\|\zeta'\|$ $q' = \zeta'/\bar{h} - \bar{\zeta}\eta'/\bar{h}^2$: how much vortex stretching,

$\|\eta'\|/\|\boldsymbol{u}'\|$ should scale as F^2 for a balanced mode,

$T_{\text{shear}} = -\int u_s' u_n' \, \partial_n \bar{U}_s$ the ONLY channel by which the mean-flow PV gradient can power a wave,

$T_{\text{bend}} = -\int u_n' \, \bar{U}_s^2 \, \partial_s^2 \zeta_c$ work done on the fluid by the moving bank.

The decisive one is T_{shear}. A free vortical (Rossby-type) wave is *defined* by extracting wave activity from the background PV gradient, so it requires $T_{\text{shear}} > 0$. If instead $T_{\text{shear}} \le 0$ the mean flow is a *sink* and the disturbance is being driven by the boundary, whatever its Froude behaviour. Note also that a smooth, single-signed $\partial_n \bar{q}$ (which is what a parabolic jet gives) has no PV edge and therefore supports *no discrete* vortical eigenmode at all — only a continuous spectrum [15]; discrete counter-propagating Rossby waves require localised PV gradients [16]. Planetary β supports Rossby waves with no mean flow whatever; a shear-β does not, because it *is* the mean flow.

14 What the parameter study actually returns

`experiments.py` runs one broadband simulation per physical configuration, varying the bed, the initial bank sinuosity, the friction and the two base speeds one at a time. Table 1 is generated directly from the output files by `make_results_table.py` — it is never typed by hand, so it cannot drift from the data it describes.

Four readings follow. The first kills the headline one might have hoped for; the second and third are corrections to earlier versions of this section that the controls forced, and are recorded as such rather than quietly rewritten.

The mean flow is a sink, not the power source. Across all eleven runs the correlation is exact: **every growing run has $T_{\text{shear}} \le 0$, and the single run with $T_{\text{shear}} > 0$ is the one that decays** (the matched reversed-shear control, $T_{\text{shear}}/|T_{\text{bend}}| = +0.42$, $\sigma(k{=}1.8) = -0.053$). Whenever this disturbance grows it is *losing* energy to the background vorticity gradient rather than extracting it, so by the criterion of §13 it is not a free vortical (Rossby-type) wave; the energy comes from T_{bend}, the work done on the fluid by the moving bank. Only the *sign* of this ratio should be quoted: its magnitude is resolution-sensitive (the reference run gives -3.16 at $N_s = 64$ and -0.36 at $N_s = 128$), because the diagnostic is evaluated on the final field and the resolved mode mix changes with N_s.

The channel-β is not necessary — but it is not irrelevant either, and an earlier version of this section got that wrong. The profile $\bar{U}_s = U_0 + \Delta(1 - n^2/b^2)$ has only two free constants, so removing Δ must change either the bank speed U_0 or the centre speed: *no perfectly clean control exists*. The first attempt varied Δ while also raising U_0 from 0.4 to 1.0, and since the erosion law reads u_s *at the bank*, that conflated "removed the channel-β" with "sped up the bank". The confound

bed H	bank	C_f	U_0	Δ	$\sigma(k{=}1.8)$	$k_{\max}$	$\sigma_{\max}$	c	$\|\delta'\|/\|\zeta'\|$	$T_{\text{shear}}/\|T_{\text{bend}}\|$
0.3	0.018	0.05	0.4	+0.6	+0.1805	1.88	+0.1815	+0.659	0.0517	-0.379
flat	0	0.05	0.4	+0.6	+0.1651	1.80	+0.1651	+0.666	0.0099	-0.312
flat	0.018	0.01	0.4	+0.6	+0.1895	1.80	+0.1895	+0.671	0.0100	-0.337
flat	0.018	0.05	0.4	+0	+0.0599	$4.72^\dagger$	$+0.1506^\dagger$	+0.405	0.0007	-0.000
flat	0.018	0.05	0.4	+0.6	+0.1649	1.80	+0.1649	+0.666	0.0099	-0.311
flat	0.018	0.05	0.4	-0.3	-0.0526	*stable*	*stable*	*stable*	0.0966	+0.405
flat	0.018	0.05	0.8	+0.2	+0.4273	$4.72^\dagger$	$+1.0446^\dagger$	+0.056	0.0105	-2.856
flat	0.018	0.05	1	+0	+0.5441	$4.72^\dagger$	$+1.4867^\dagger$	+0.216	0.0177	-0.000
flat	0.018	0.05	1	-0.6	+0.2192	$4.72^\dagger$	$+0.9042^\dagger$	+0.008	0.0139	-179.554
flat	0.018	0.1	0.4	+0.6	+0.1334	1.88	+0.1338	+0.665	0.0098	-0.285
flat	0.15	0.05	0.4	+0.6	+0.1540	1.80	+0.1540	+0.670	0.0103	-0.293

† 4 run(s): $\sigma(k)$ is still *rising* at the last resolved wavenumber, so $k_{\max}$ and $\sigma_{\max}$ are set by N_s and not by the equations (they moved from $k{=}2.32, \sigma{=}0.74$ to $k{=}4.72, \sigma{=}1.49$ when N_s was doubled). Compare these rows only through the fixed-k column.

Table 1: Fastest *converged* mode (≥ 3 e-foldings) per configuration, and the energetics ratio. $T_{\text{shear}} > 0$ would mean the mean-flow vorticity gradient powers the disturbance.

inverts the answer. Holding the bank speed fixed at the reference value and varying only Δ, compared at the fixed well-resolved wavenumber $k = 1.8$:

Δ	$\sigma(k{=}1.8)$	
+0.60 (reference)	+0.1649	grows
+0.00 (β removed)	+0.0599	grows, but 2.75× *slower*
−0.30 (reversed)	−0.0526	**stable** (no mode converged)

What survives is the qualitative statement: with the background vorticity gradient *identically zero* the meander still grows ($\sigma > 0$), so the mechanism does not require β and cannot be a β-restored wave. What does *not* survive is the claim that it grows *faster* without β — that was the U_0 confound. Removing β weakens the instability by a factor $\simeq 2.75$, and reversing the shear at matched bank speed suppresses it altogether.

Wavelength selection: there is one, and it comes from the shear. This too corrects an earlier reading. Fitting $\sigma \simeq ak - bk^2$ gives $k_{\text{peak}} = a/2b$, and the measured damping coefficient is $b \approx 0.079$ while $\nu = 0.003$ — the k^2 damping that produces the peak is some **twenty-six times larger than the explicit viscosity**, so the cutoff is not viscous. It tracks the shear instead: $b = 0.079, 0.038, 0.033, -0.003$ for $\Delta = 0.6, 0.2, 0.0, -0.6$ (reversed shear *anti*-damps), and the peak obeys

$$\boxed{k_{\text{peak}}\,\Delta \simeq 1.10} \tag{21}$$

to within ±3% across a tenfold change in C_f, a flat-to-parabolic bed, and bank sinuosity $0 \to 0.15$. The earlier argument — "$k_{\max}$ barely moves while C_f changes tenfold, therefore it is the ν cutoff" — was under-determined, because all six of those runs held $\Delta = 0.6$; "Δ-set" fits the same evidence, and varying Δ discriminates. As $\Delta \to 0$ the shear damping vanishes and the peak retreats to a residual viscous-plus-drag cutoff ($b_0 \approx 0.011$, $k_{\text{peak}} = 4.30$). So the model does select a length scale, but by cross-channel shear, not by the bar–bend resonance of §12(i); a comparison with the observed $\lambda \approx 10W$ remains unwarranted.

Why the movies look untidy, and why that is not numerical noise. The perturbation field never settles into a single clean travelling wave, and it is worth being precise about why, because the obvious reading — "the simulation is noisy" — is wrong on both counts.

It is a wave packet, not noise. Measured on the final field, 90% of the streamwise-velocity variance lives in about **seven neighbouring modes**, $k \simeq 1.65$–2.10, an r.m.s. spectral width of only $\Delta k \approx 0.15$. Seven nearby wavenumbers with unequal phases beat against one another, which looks irregular but is entirely deterministic; broadband noise would instead show a flat spectrum. The packet width is even predictable: near its maximum $\sigma \simeq \sigma_{\max} - a(k-k_p)^2$ with $a \approx 0.077$ measured from the table, so after a time T only modes with $a(k-k_p)^2 T \lesssim 1$ survive, i.e. $\Delta k \sim (aT)^{-1/2} \approx 0.3$ at $T = 120$ — consistent with what is measured. The packet therefore narrows only as $T^{-1/2}$, and the `max_efold` cap stops the run long before it becomes monochromatic.

The grid is clean. The energy in the top decade of resolved wavenumbers is $\sim 10^{-16}$ of the total: there is no aliasing pile-up at all. And refining N_s from 64 to 128 at fixed ν changes $\sigma(k)$ by 0.0–0.1% across the whole band, leaving the peak at $k = 1.80$ — the structure is converged, not grid-generated.

What $N_s = 128$ does fix is legibility. The dominant wavelength is $\lambda = 2\pi/1.86 \approx 3.4$, which at $N_s = 64$ is only 2.6 grid points per wavelength — barely above the two-point Nyquist limit, so *any* wave, however clean, would render as one-pixel stripes. At $N_s = 128$ it is 5.2 points per wavelength and draws as the wave it is. The production runs use $N_s = 128$ for that reason alone; the physics is unchanged.

A note on run length, because it sets the numbers. Since all modes are seeded together, two of them separated by $\Delta\sigma$ reach an amplitude ratio $e^{\Delta\sigma T}$; once that exceeds $1/\varepsilon \approx 4.5 \times 10^{15}$ (some 36 e-foldings) the slower one is buried in the faster one's FFT round-off and simply reads back the *leader's* growth rate. The e-folding gate of §10 cannot catch this, because round-off inherits the leader's growth and therefore looks perfectly converged; it is visible instead as a spurious sawtooth in $\sigma(k)$, in which σ depends only on m modulo a fixed integer. At a common $t_{\text{end}} = 120$ the slowest configuration here grew a harmless 13 e-foldings while the plug-flow run grew 58, and a third of the latter's spectrum was noise. Runs are therefore stopped when the *leading* mode has grown `max_efold`= 25 e-foldings, or at t_{end}, whichever comes first ($\rightarrow$ `run_ivp_SW`): the six slow runs are unaffected and the three fast ones stop at $t = 36$–80. This ties the usable dynamic range to the diagnostic rather than to how fast a configuration happens to grow, and it is why the fast runs are integrated over a shorter interval than the slow ones.

What the model *is*, then. A forced morphodynamic boundary mode: balanced — the divergence ratio is $\|\delta'\|/\|\zeta'\| \simeq 0.010$–$0.051$, and F barely matters, both consistent with the rigid-lid reduction — but driven by the erodible boundary rather than by the interior PV gradient. The growth rate vanishes with the erodibility, $\sigma \propto E$: it is not an eigenvalue of the flow alone. The cross-channel shear then shapes it, setting both the amplitude (a factor 2.75 between $\Delta = 0.6$ and $\Delta = 0$, and outright stability for $\Delta < 0$ at matched bank speed) and the selected wavenumber ($k_{\text{peak}}\Delta \simeq 1.10$) — but it never *powers* it, since $T_{\text{shear}} \leq 0$ in every growing run. The useful corollary is a validation result: since the free surface is passive here, the rigid-lid packages in `../` are justified for $F \lesssim 0.9$.

A methodological note worth keeping. Two of the four readings above are corrections, and both were caught the same way — not by re-deriving, but by refining a numerical parameter and by adding a control that held fixed the variable which had been riding along. The U_0/Δ confound was invisible at $N_s = 64$ because the runs it corrupted were exactly the runs whose $\sigma(k)$ had not converged; doubling N_s made their growth rates move ($\sigma_{\max}$: $0.74 \rightarrow 1.49$), which is what prompted the control in the first place. A parameter that cannot be varied independently — here, because $\bar{U}_s$

has only two constants — should be treated as confounded by default and reported with an explicitly matched comparison, not with each run's own optimum.

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